

# Customer Behavior Analysis

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## Dataset Overview

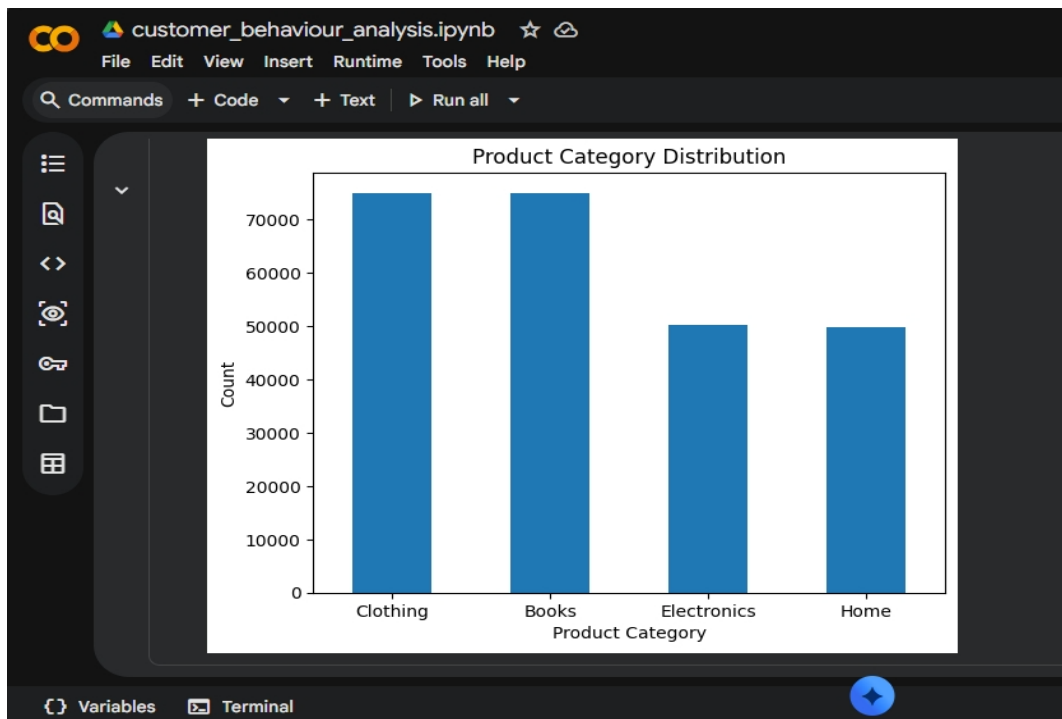
- **Dataset:** Customer Behavior Analysis Dataset
- **Tool Used:** Python, Pandas, Google Colab
- **Objective:** Analyze customer behavior and identify churn risks.

## Data Cleaning

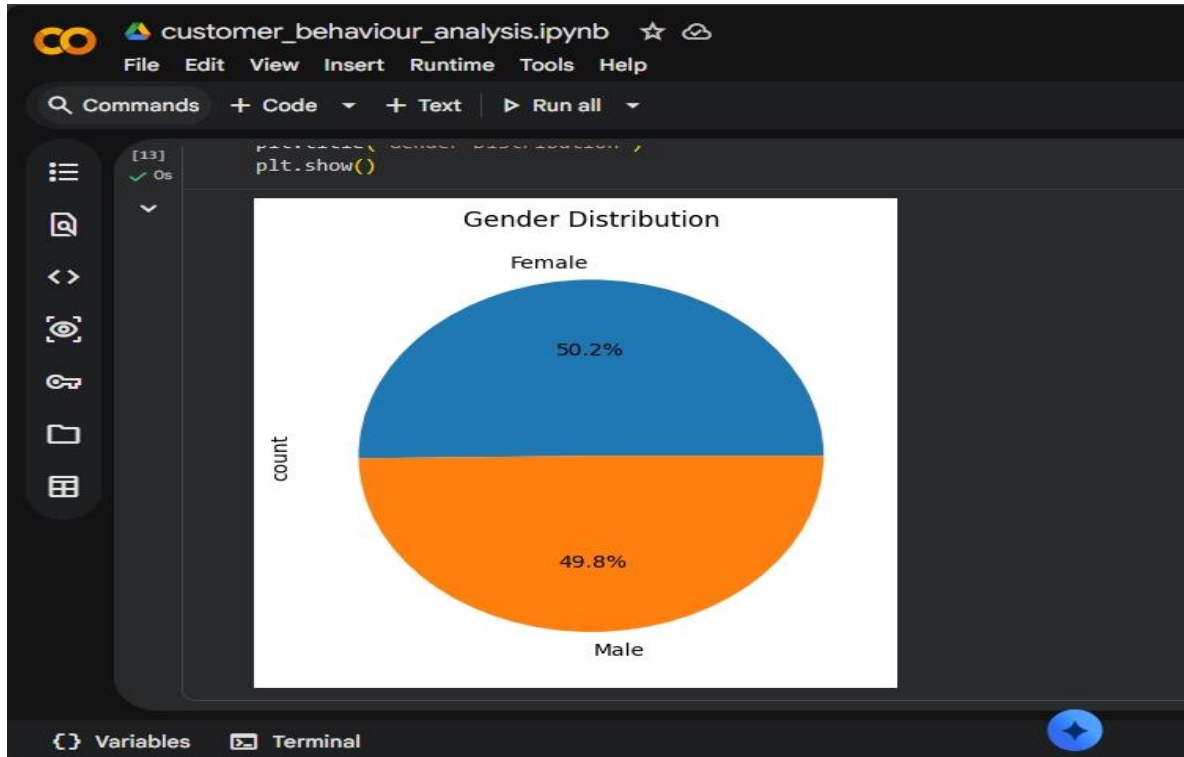
- Checked missing values.
- Removed duplicate records.
- Verified data types.

## Visualizations

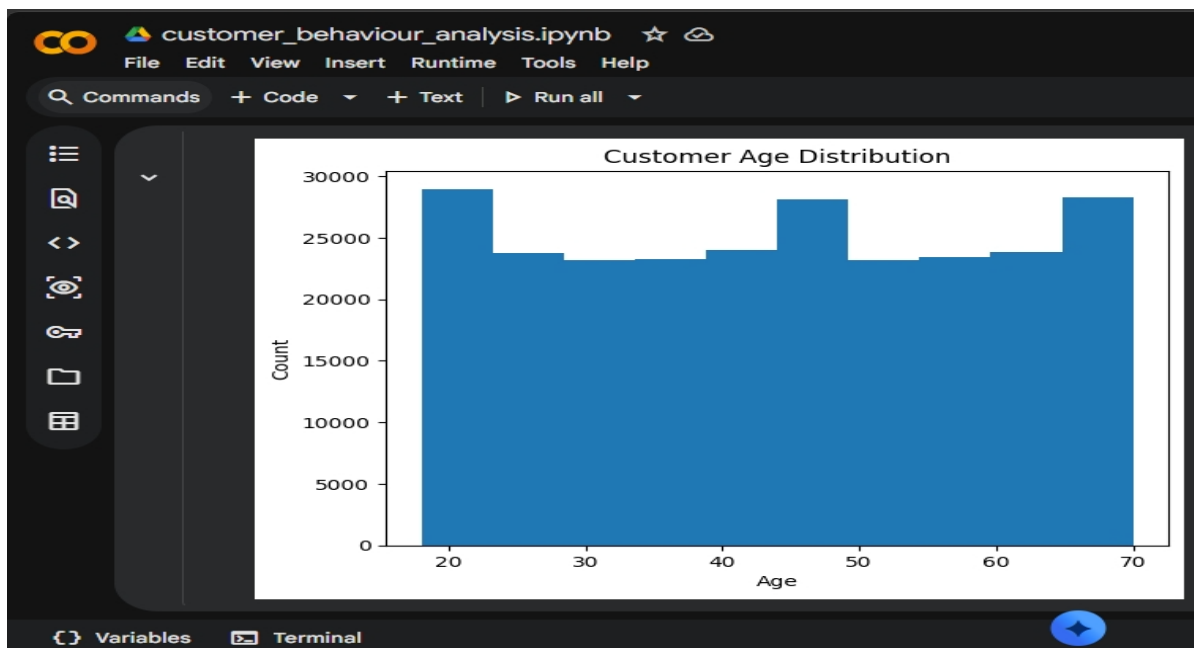
### Product Category Distribution



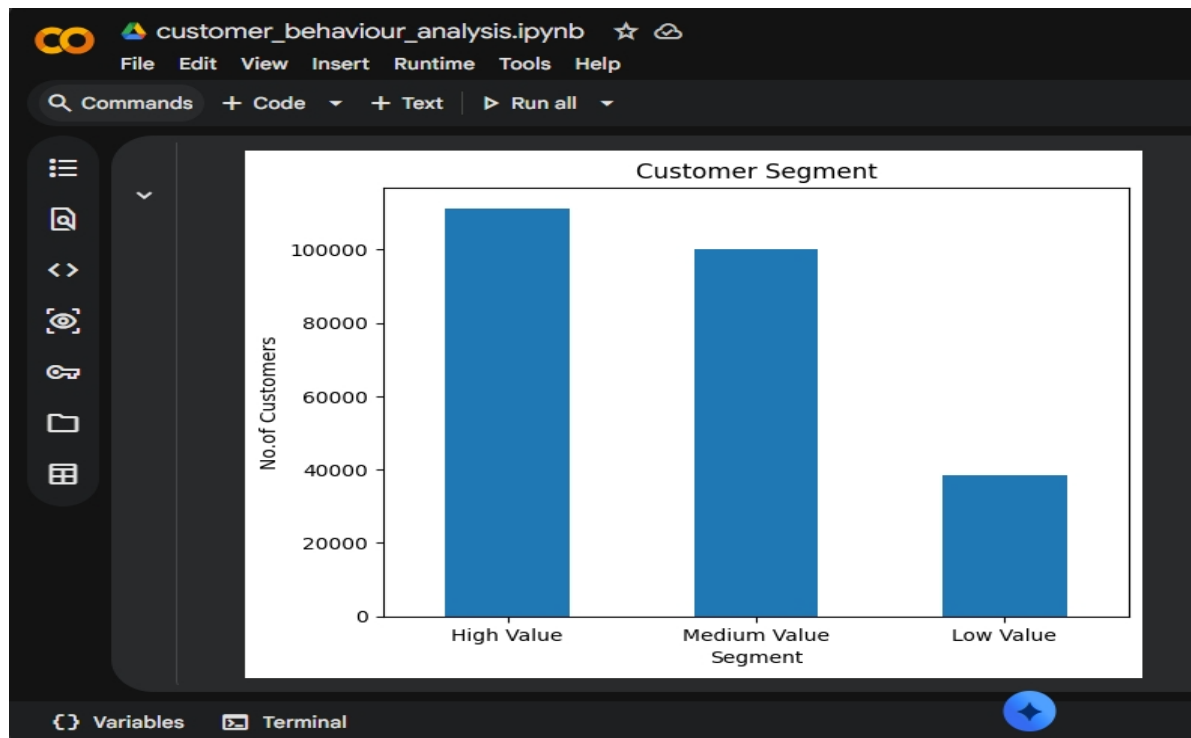
## Gender Distribution



## Customer Age Distribution



## Customer Segments



### Customer Segmentation:

- Medium Value Customers: 111,230
- High Value Customers: 100,228
- Low Value Customers: 38,542

### Churn Analysis:

- Active Customers: 200,126
- Churned Customers: 49,874

### Key Findings:

1. Medium Value customers form the largest customer segment.
2. High Value customers contribute significantly to business revenue.
3. Low Value customers are more likely to become inactive.
4. Customer retention is strong, with approximately 80% active customers.
5. Nearly 50,000 customers have churned and require re-engagement strategies.

### Recommendations:

1. Launch loyalty programs for High Value customers.

2. Provide personalized offers to Low Value customers.
3. Create upselling campaigns for Medium Value customers.
4. Implement churn prediction and customer retention initiatives.
5. Use customer purchase history for personalized recommendations.

## Conclusion:

Customer behavior analysis identified valuable customer segments and churn risks. The findings can help Alfido Tech improve customer retention and engagement strategies.